```
#include <iostream>

#include <chrono>


using namespace std;

using namespace chrono;


unsigned long long factorialIterative(int n)

{

    unsigned long long result = 1;


    for (int i = 1; i <= n; i++)

    {

        result = result * i;

    }


    return result;

}


unsigned long long factorialRecursive(int n)

{

    if (n == 0 || n == 1)

    {

        return 1;

    }


    return n * factorialRecursive(n - 1);

}


int main()

{

    int n;
```

```
cout << "Enter a non-negative integer: ";
cin >> n;
if (n < 0)
{
    cout << "Invalid input! Please enter a non-negative integer." << endl;
    return 1;
}

auto startIterative = high_resolution_clock::now();

unsigned long long iterativeResult = factorialIterative(n);

auto endIterative = high_resolution_clock::now();

auto iterativeTime =
    duration_cast<nanoseconds>(endIterative - startIterative);


auto startRecursive = high_resolution_clock::now();

unsigned long long recursiveResult = factorialRecursive(n);

auto endRecursive = high_resolution_clock::now();

auto recursiveTime =
    duration_cast<nanoseconds>(endRecursive - startRecursive);


// ---------------- DISPLAY RESULTS ----------------
```

```
    cout << "\n========== FACTORIAL RESULTS ==========" << endl;

    cout << "Number              : " << n << endl;

    cout << "\n--- Iterative Method ---" << endl;
    cout << "Factorial Result      : " << iterativeResult << endl;
    cout << "Execution Time        : "
         << iterativeTime.count() << " nanoseconds" << endl;

    cout << "\n--- Recursive Method ---" << endl;
    cout << "Factorial Result      : " << recursiveResult << endl;
    cout << "Execution Time        : "
         << recursiveTime.count() << " nanoseconds" << endl;

    cout << "\n=================" << endl;

    return 0;
}
```